EXCELLENCE

CLASS 11 - 2026-27

Physics | Units 06

Units and Measurement

Easy Instruments: Scale, Vernier, Screw Gauge

By - Excellence Programme

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What we learn today

01 Metre scale — the simple ruler

02 Vernier callipers — how to find least count

03 How to read a vernier

04 Zero error — when the instrument is wrong at zero

05 Screw gauge — for very small lengths

Link from last class

Last class we learned how error grows when we add, multiply or raise a number to a power. That part is done.

Today we learn the tools that measure length: a metre scale, vernier callipers and a screw gauge. The key word is least count — the smallest length the tool can measure.



Least count = smallest length the instrument can show

Class notes come after teaching. These slides are for the board.

Short names we will use

- L.C. = least count (smallest step the tool can show)
- M.S.D. = one small mark gap on the main scale
- V.S.D. = one small mark gap on the vernier scale
- M.S.R. = main scale reading | V.S.R. = vernier scale reading

1 cm, 1 mm

Learn these four short names. Questions use them a lot.

Three tools for length

- Metre scale — a normal ruler with millimetre marks.
- Vernier callipers — a ruler plus a short sliding scale.
- Screw gauge — a screw tool for very thin objects.

All three measure length. Least count tells how fine the reading can be.

Rough work / extra work

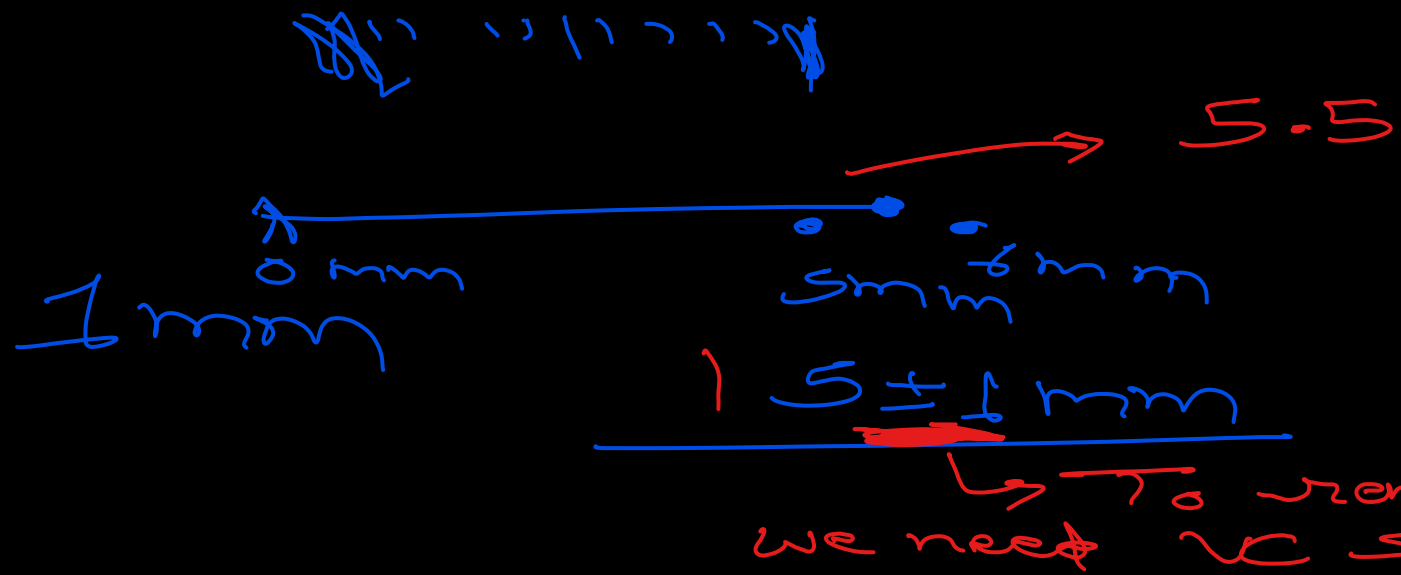
Metre scale

On a school metre scale, marks are 1 mm apart. So the smallest length you can read is 1 mm. That 1 mm is the least count.

Least count of metre scale = 1 mm

Write the answer as (reading ± 1 mm)

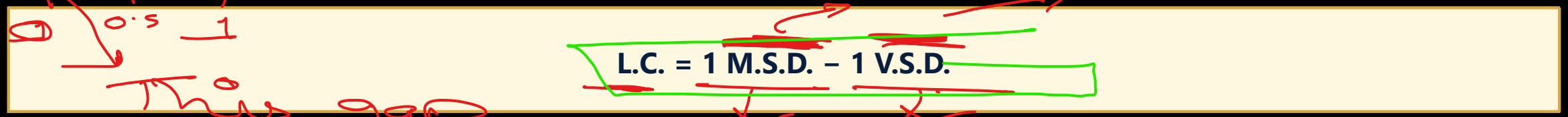
If a rod ends between 4 mm and 5 mm, write 4 mm ± 1 mm. Do not guess 4.2 mm.



Why we need a vernier

A metre scale cannot show lengths smaller than 1 mm. Vernier callipers fix that. They have a fixed main scale and a short scale that slides. That short scale is the vernier.

The vernier marks are a little closer than the main marks. So one vernier mark can line up exactly with a main mark. That helps us read a small part of 1 mm.



Easy school vernier

In school, 10 vernier marks match 9 main marks. Each main mark gap is 1 mm. So one vernier gap is a bit less than 1 mm.

10 V.S.D. = 9 mm → 1 V.S.D. = 0.9 mm

L.C. = 1 mm − 0.9 mm = 0.1 mm

If n vernier marks match (n − 1) main marks: L.C. = (1 M.S.D.) / n

Only two jobs

- Job 1: Find the least count from how the two scales match.
- Job 2: Find the reading = main reading + (vernier mark × least count).

Reading = M.S.R. + (V.S.R. × L.C.)

Write the final answer as (reading ± least count).

Least count: 9 and 12

1 msd = 1 cm

Solve

One main mark gap is 1 cm. 9 main gaps match 12 vernier gaps. Find the least count.

VSD

12 vsd = 9 mm

12 vsd = 9 cm

1 vd = $\frac{9}{12}$ cm

= $\frac{3}{4}$ ⇒ 0.75 cm

1 vsd = 0.75

LC = msd − vsd

= 1 cm − 0.75 cm

= 0.25 cm

Solution: Least count: 9 and 12

Idea: find one vernier gap, then subtract it from one main gap.

Step 1: 1 V.S.D. = (9/12) × 1 cm = 0.75 cm

Step 2: L.C. = 1 − 0.75 = 0.25 cm

Final answer: 0.25 cm

Rough work / extra work

25 vernier marks

1 msd = 1 mm

25 vernier marks match 24 main marks. One main mark gap is 1 mm. Least count is

25 vsd = 24 msd

1 vsd = $\frac{24}{25}$ mm

1. ~~0.04 mm~~
2. 0.01 mm
3. 0.02 mm
4. ~~0.08 mm~~

LC = 1 msd − 1 vsd

= 1 mm − 0.96 mm

= 0.04 mm

$\frac{96}{100}$ mm

0.96 mm

Solution: 25 vernier marks

Idea: use L.C. = 1 M.S.D. − 1 V.S.D.

Step 1: 1 V.S.D. = 24/25 mm = 0.96 mm

Step 2: L.C. = 1 − 0.96 = 0.04 mm

Final answer: 0.04 mm (option 1)

Formula with x and n

Same If

One main mark gap is x cm. n vernier marks match (n – 1) main marks. Least count in cm is

1. (n – 1)x / n
2. n x / (n – 1)
3. ~~x / n~~
4. x / (n – 1)

Solution: Formula with x and n

Idea: this is the same school rule, written with letters.

Step 1: 1 V.S.D. = ((n − 1)/n) × x

Step 2: L.C. = x − ((n − 1)/n)x = x/n

Final answer: x / n (option 3)

Rough work / extra work

20 vernier and 8 main

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1 cm on the main scale is cut into 10 equal parts. 20 vernier marks match 8 of those small parts. Find least count.



Solution: 20 vernier and 8 main

Idea: find 1 M.S.D., then 1 V.S.D., then subtract.

Step 1: 1 M.S.D. = 1/10 cm = 0.1 cm

Step 2: 1 V.S.D. = (8/20) × 0.1 = 0.04 cm

Step 3: L.C. = 0.1 − 0.04 = 0.06 cm

Final answer: 0.06 cm

How to read a vernier

Look at the zero mark of the vernier. The main mark just before it is the main scale reading. Then look for the vernier mark that lines up with any main mark. That number is the vernier reading.

Reading = M.S.R. + V.S.R. × L.C.

Example: 3 mm + 4 × 0.1 mm = 3.4 mm

Write it as (3.4 ± 0.1) mm

What is zero error?

Close the jaws with nothing inside. The two zero marks should meet. If they do not meet, the tool already shows a small wrong reading. That wrong reading is called zero error.

Always subtract the zero error from the reading you took. Keep the + or − sign. Zero correction is just the opposite of zero error.

True reading = reading you took − zero error

Positive zero error

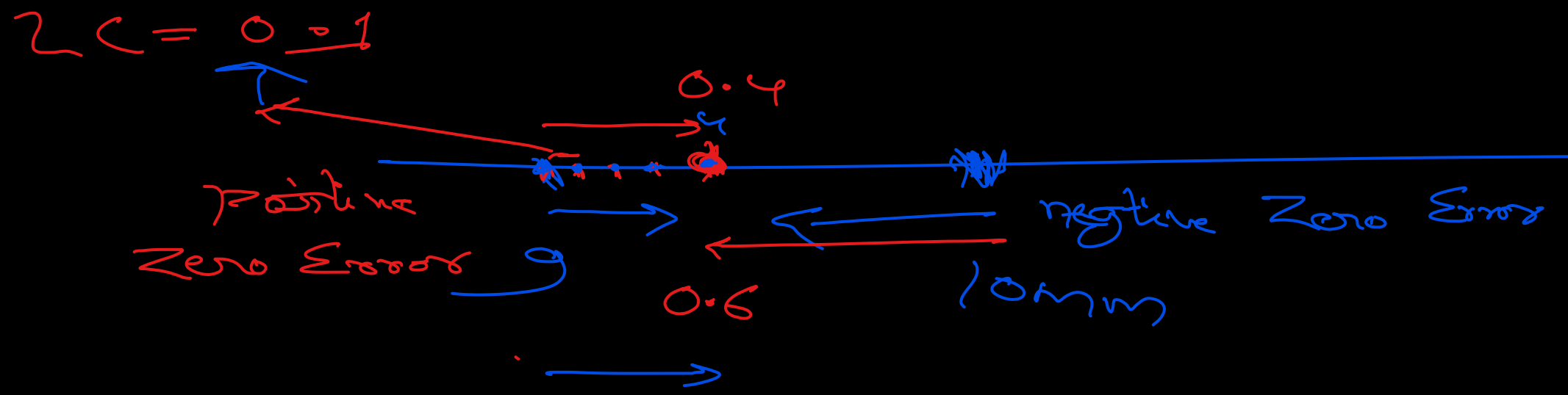
Jaws closed. If vernier zero is to the right of main zero, zero error is positive. Size = (matching vernier mark) × least count.

Zero error = + (matching vernier mark) × L.C.

Example: 4th mark matches, L.C. = 0.1 mm → zero error = +0.4 mm

Zero correction here is −0.4 mm. Error and correction have opposite signs.

Rough work / extra work



4 × 0.1 = 0.4

(10 − 4) × 0.1

= − 6 × 0.1 ⇒ − 0.6

Negative zero error

Jaws closed. If vernier zero is to the left of main zero, zero error is negative. For a 10-mark vernier, use (10 − matching mark) × least count, and put a minus sign.

Zero error = − (N − matching mark) × L.C.

Example: N = 10, 6th mark matches, L.C. = 0.1 mm → zero error = −0.4 mm

Minus a negative error means you add. True = reading + 0.4 mm

Zero Error = − Zero Correction

Find zero correction

Jaws closed. Vernier zero is to the right of main zero. 5th vernier mark matches. L.C. = 0.1 mm. Zero correction is

1. 0.4 mm
2. +0.5 mm
3. −0.5 mm
4. −0.4 mm

Zero Error = 5 × 0.1 = 0.5

Zero Correction = − (Zero Error)

= − 0.5

= − 0.5

Solution: Find zero correction

Idea: right side means + error. Correction flips the sign.

Step 1: Zero error = +5 × 0.1 = +0.5 mm

Step 2: Zero correction = −0.5 mm

Final answer: −0.5 mm (option 3)

Fix a vernier reading

0.02 cm

Zero error = +0.2 mm. Vernier zero is between 8.5 cm and 8.6 cm. Matching vernier mark = 6. L.C. = 0.01 cm. Correct length is

1. 8.58 cm
2. 8.54 cm
3. 8.56 cm
4. 8.36 cm

8.5 + 6 × 0.01

8.56

Remove Zero Error

8.56 − 0.02

8.54 cm

Solution: Fix a vernier reading

Idea: use one unit. Change 0.2 mm to 0.02 cm.

Step 1: Reading taken = 8.5 + 6 × 0.01 = 8.56 cm

Step 2: True length = 8.56 − 0.02 = 8.54 cm

Final answer: 8.54 cm (option 2)

Rough work / extra work

Sphere with zero error

$LC = \frac{1 \text{ msd}}{n} \Rightarrow \frac{0.1}{10} = 0.01$

1 M.S.D. = 0.1 cm. 10 vernier marks match 9 main marks. Jaws closed: 6th vernier mark matches (vernier zero is left). With sphere: M.S.R. = 3.1 cm, 1st vernier mark matches. Diameter is

1. 3.07 cm
2. 3.11 cm
3. 3.15 cm
4. 3.17 cm

10 vsd = 9 msd

1 vsd = 0.9 msd

1 vsd = 0.09 cm

LC = 0.01

Negative Zero Error

$-(10 - 6) \times LC$

-4×0.01

$= -0.04$

Reading = MSR + VSR × LC

$= 3.1 + 10 \times 0.01$

$= 3.11$

Then True Reading = Reading − Zero Error

$= 3.11 - (-0.04) = 3.15$

Solution: Sphere with zero error

Idea: least count → zero error → sphere reading → true value.

Step 1: L.C. = 0.1 / 10 = 0.01 cm

Step 2: Zero error = −(10 − 6) × 0.01 = −0.04 cm

Step 3: Reading taken = 3.1 + 1 × 0.01 = 3.11 cm

Step 4: True = 3.11 − (−0.04) = 3.15 cm

Final answer: 3.15 cm (option 3)

Diameter of a cylinder

msd = 0.05

Solve

No zero error. Vernier zero is between 5.10 cm and 5.15 cm. 50 vernier marks = 2.45 cm. 24th vernier mark matches. Diameter is

1. 5.112 cm
2. 5.124 cm
3. 5.136 cm
4. 5.148 cm

Solution: Diameter of a cylinder

Idea: gap 5.10 to 5.15 means 1 M.S.D. = 0.05 cm.

Step 1: 1 V.S.D. = 2.45 / 50 = 0.049 cm

Step 2: L.C. = 0.05 − 0.049 = 0.001 cm

Step 3: Diameter = 5.10 + 24 × 0.001 = 5.124 cm

Final answer: 5.124 cm (option 2)

20 vernier match 16 main

Main scale marks are 1 mm apart. 20 vernier marks match 16 main marks. Least count is

1. 0.02 mm
2. 0.05 mm
3. 0.1 mm
4. 0.2 mm

Solution: 20 vernier match 16 main

Idea: same subtract rule again.

Step 1: 1 V.S.D. = 16/20 = 0.8 mm

Step 2: L.C. = 1 − 0.8 = 0.2 mm

Final answer: 0.2 mm (option 4)

Rough work / extra work

n marks in one centimetre

How

Main scale has n marks in 1 cm. n vernier marks match (n – 1) main marks. Least count is

1. 1 / ((n + 1)(n – 1)) cm
2. 1 / n cm
3. 1 / n^2 cm
4. 1 / (n(n + 1)) cm

Solution: n marks in one centimetre

Idea: one main gap is already 1/n cm.

Step 1: 1 M.S.D. = 1/n cm

Step 2: L.C. = (1 M.S.D.) / n = $1/n^2$ cm

Final answer: $1 / n^2$ cm (option 3)

Fill the blank

Vernier constant = 0.1 mm. Zero error = (−0.05) cm. M.S.R. = 1.7 cm. Matching vernier mark = 5.
Correct diameter = ____ × 10^{-2} cm.

Solution: Fill the blank

Idea: change 0.1 mm to cm first, then find true length.

Step 1: L.C. = 0.1 mm = 0.01 cm

Step 2: Reading taken = 1.7 + 5 × 0.01 = 1.75 cm

Step 3: True = 1.75 − (−0.05) = 1.80 cm = 180×10^{-2} cm

Final answer: 180

Rough work / extra work

Least count in mm

Solve it

One main mark gap is a cm. The nth vernier mark matches the (n – 1)th main mark. Least count in mm is

1. 10 n a / (n – 1)
2. 10 a / (n – 1)
3. 10 a / n
4. (n – 1)a / (10 n)

Good

Solution: Least count in mm

Idea: in cm, L.C. = a/n. Times 10 gives mm.

Step 1: L.C. = a/n cm = 10a/n mm

Final answer: 10 a / n (option 3)

Screw gauge in plain words

A screw gauge is used for very small lengths, like the diameter of a thin wire. When you turn the screw once, it moves forward by a fixed length. That fixed length is called pitch.

→ similar to V.S.D

Pitch = how far the screw moves in one full turn

L.C. = Pitch / (number of marks on the round scale) → Pitch = 2 (x) no. of marks on Round scale

Reading = linear reading + (round-scale mark × L.C.) → Remember

→ Zero Error

Zero error of screw gauge

Close the ends with nothing between them. The zero of the round scale should meet the line. If it does not, there is a zero error — just like in the vernier.

True reading = reading you took − zero error

Screw gauge least count is smaller than vernier. That is why we use it for thin wires.

Find the pitch

A screw gauge has least count 0.01 mm and 50 marks on the round scale. Pitch is

1. 0.01 mm
2. 0.25 mm
3. 0.5 mm
4. 1.0 mm

Solution: Find the pitch

Idea: Pitch = least count × number of marks.

Step 1: Pitch = 0.01 × 50 = 0.5 mm

Final answer: 0.5 mm (option 3)

Rough work / extra work

Wire diameter

Pitch = 0.5 mm. Round scale has 50 marks. Linear reading = 2.5 mm. Round reading = 20. Zero error = +0.02 mm. True diameter is

$$LC = \frac{\text{Pitch}}{\text{No. of Round}} = \frac{0.5}{50} = 0.01$$

$$\text{Reading} = 2.5 + (20 \times 0.01) \Rightarrow 2.5 + 0.2$$

$$TR = R - ZE$$

$$= 2.7 - 0.02 \quad 2.5 + 0.2$$

$$= 2.68 \qquad 2.7$$

Solution: Wire diameter

Idea: least count → reading taken → subtract zero error.

Step 1: L.C. = 0.5 / 50 = 0.01 mm

Step 2: Reading taken = 2.5 + 20 × 0.01 = 2.70 mm

Step 3: True = 2.70 − 0.02 = 2.68 mm

Final answer: 2.68 mm

Remember these lines

Metre scale least count = 1 mm

Vernier: L.C. = 1 M.S.D. − 1 V.S.D. (or 1 M.S.D. / n)

Reading = M.S.R. + V.S.R. × L.C.

True reading = reading taken − zero error

Screw gauge: L.C. = Pitch / N. Same zero-error rule.

Four lines to write

1. Vernier least count = 1 M.S.D. − 1 V.S.D.

2. Reading taken = M.S.R. + (matching vernier mark × L.C.)

3. True reading = reading taken − zero error

4. Screw gauge least count = Pitch / (round marks)

These four lines are enough for most board questions.

Rough work / extra work

Try these at home

1. 50 vernier marks match 49 main marks. 1 M.S.D. = 1 mm. Find least count.

2. M.S.R. = 4.2 cm, V.S.R. = 7, L.C. = 0.01 cm, zero error = +0.03 cm. Find true length.

3. Pitch = 1 mm, round scale has 100 marks. Find least count in mm and in cm.

4. Ends closed. Round reading = −3, L.C. = 0.01 mm. Write zero error and zero correction.

Units chapter is finished. Next chapter: Motion in a Straight Line.